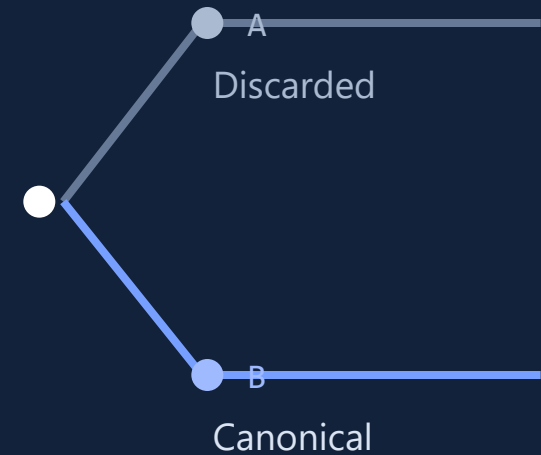


Same height. Different history.

Reproduce the events your projection kept after the chain discarded them.

REAL BYTECODE. LOCAL DEVELOPMENT EVM.



SAME BLOCK NUMBER

A correct total can hide four wrong task states.

COMPLETED TASKS

2

=

2

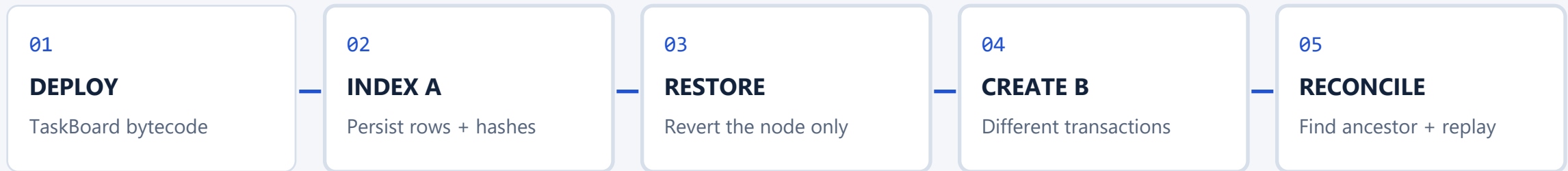
Same total.

The height-only cursor preserves two orphaned events and misses two canonical events.

TASK	STALE PROJECTION	CONTRACT ON B
#7	Open	Complete
#9	Open	Complete
#42	Complete	Open
#99	Complete	Open

Observed in `rl-1e4e5f8aae75`

Replace only the chain. Preserve the stale index.



INDEPENDENT REFERENCE

Contract storage is read with `eth_call` at a pinned canonical block hash. Canonical log identities are checked as well.

Hardhat / EDR
Solidity / ethers
TypeScript / JSON

A and B reach block 3; their hashes differ. The saved index stays unchanged during restore.

Prove recovery. Report uncertainty.

7 positive checks observed

- + Normal indexing
- + Branch replacement exercised
- + Expose the flawed projection
- + Recover state and exact rows
- + Replay without duplicate rows
- + Restart from stale disk state
- + Second projector example

2 expected inconclusive results

Insufficient rollback depth: reconstruct before claiming recovery.

Interrupted RPC read: discard the candidate snapshot; keep stored data unchanged.

These are scoped fixture controls, not a security score or proof about every chain reorganization.

A small harness. An honest boundary.

WHAT YOU CAN ADAPT

A typed projection interface, memory and JSON stores, a fresh local chain, and a portable report.

The included second projector counts completion transitions and checks its answer against contract storage.

No mainnet, funds, consensus testing, universal repair or production security certification.

WHAT ALREADY EXISTS

Ponder, SQD and Etherfold already handle reorganizations. portal-ponder is a close testing precedent.

Our contribution is the bounded experiment and explanation. It does not plug into any existing indexer without integration work.

Run it.

Inspect the difference.

NODE 24 / FROM A CLEAN CHECKOUT

```
npm ci  
npm run lab  
npm run dev
```

AI-assisted development with Codex. Original experiment built for this event.

Report: [r1-1e4e5f8aae75](#) / 2026-09-13

SOURCE + INSTRUCTIONS

github.com/EazyHood/reorg-lab

The web viewer loads a saved run. Execute locally and import its JSON to inspect your own result.